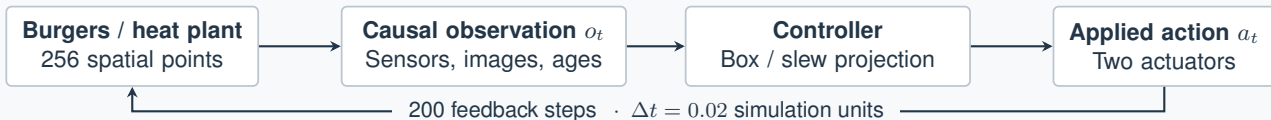
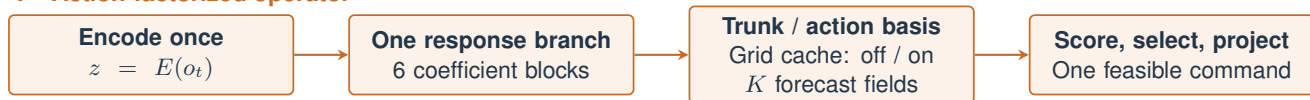


(a) Closed-loop control

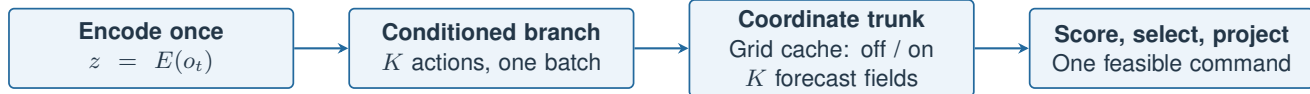


(b) Learned predictors and reference-solver candidate controllers

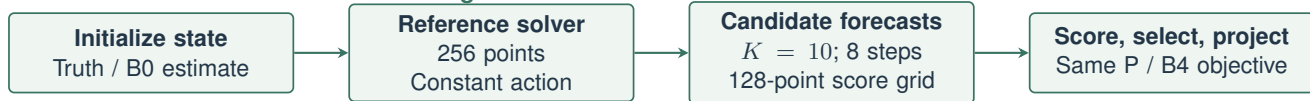
P Action-factorized operator



B4 Shared-encoder, candidate-batched operator



RS-true / RS-obs Reference-solver diagnostic



Control: $K = 10$, eight-step horizon. **Timing:** $K = 10, 25, 50, 100, 200$; cache off / on.

Fixed-grid trunk and spatial-basis caches are specific to each frozen model; encoding remains once per request.

(c) Primary evaluation / Follow-up diagnostics

